THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A triangle has sides 14 cm, 48 cm and 50 cm. It is: 1

(a) equilateral **(b)** not right-angled **(c)** right-angled **(d)** cannot be said

2. In a survey of 60 students, 14 chose cricket. In a pie chart, the central angle of the cricket sector (in degrees) is: 1

(a) 168 **(b)** 84 **(c)** 42 **(d)** 94

3. A cube of side 3 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 8 **(b)** 1 **(c)** 6 **(d)** 12

4. A plant's height (cm) at the end of weeks 1 to 6 is 4, 8, 15, 19, 22, 28. Between which two consecutive weeks was the growth the greatest? 1

(a) Between week 2 and week 3 **(b)** Between week 3 and week 4
(c) Between week 1 and week 2 **(d)** Between week 4 and week 5

5. In a calendar the nine dates in a 3×3 block have the middle date 10. Their sum is: 1

(a) 81 **(b)** 30 **(c)** 99 **(d)** 90

6. The cost of flooring a 18 m by 12 m hall at ₹ 45 per square metre is (in ₹): 1

(a) 2700 **(b)** 9720 **(c)** 261 **(d)** 1350

7. For m = 4, the Pythagorean triplet $(2m,\ m^2-1,\ m^2+1)$ is: 1

(a) 4, 15, 17 **(b)** 8, 14, 18 **(c)** 8, 15, 17 **(d)** 8, 17, 18

8. $\frac{1}{64}$ expressed as a power of 2 is: 1

(a) 2^{-6} **(b)** 2^{-5} **(c)** 2^{6} **(d)** 2^{-7}

9. A Maya numeral has 3 in the 20s place and 12 in the units place. Its value is: 1

(a) 48 **(b)** 42 **(c)** 72 **(d)** 15

10. In a kite ABCD with AB = AD and CB = CD, $\angle B = \angle D = 90^\circ$ and $\angle A = 70^\circ$. Then $\angle C$ is: 1

(a) 110° **(b)** 90° **(c)** 130° **(d)** 200°

11. The sum of the five consecutive numbers starting from 14 is: 1

(a) 85 **(b)** 70 **(c)** 75 **(d)** 80

12. 101×99 equals: 1

(a) 10001 **(b)** 9999 **(c)** 9998 **(d)** 10000

13. A car covers 240 km in 4 hours at a steady speed. In 7 hours it covers: 1

(a) 420 km **(b)** 480 km **(c)** 67 km **(d)** 247 km

14. The price of an article is ₹ 2000 and GST is charged at 5%. The bill amount (in ₹) is: 1

(a) 4200 **(b)** 2100 **(c)** 1890 **(d)** 2310

15. The diagonal of a square of side 3 cm (in cm) is: 1

(a) 6 **(b)** 3 **(c)** $\sqrt{19}$ **(d)** $3\sqrt{2}$

16. ₹ 2400 is divided among A, B and C in the ratio 1:3:4. B's share (in ₹) is: 1

(a) 800 **(b)** 1200 **(c)** 900 **(d)** 300

17. The net of a cylinder consists of: 1

(a) two circles and one rectangle **(b)** two circles and one triangle
(c) one circle and one sector **(d)** two rectangles and one circle

18. A dot plot shows: 1 dot at 3, 1 dot at 4, 6 dots at 5, 3 dots at 6, 9 dots at 7, 6 dots at 8. The mode of the data is: 1

(a) 5 **(b)** 3 **(c)** 4 **(d)** 7

19. The value of $8 \times 8 - 7 \times 9$ is: 1

(a) 1 **(b)** 2 **(c)** 8 **(d)** 0

20. 2 m² expressed in cm² is: 1

(a) 200000 **(b)** 200 **(c)** 2000 **(d)** 20000

21. The sum of the cubes of the first 4 natural numbers is: 1

(a) 104 **(b)** 10 **(c)** 20 **(d)** 100

22. In 7^4, the base is: 1

(a) 2401 **(b)** 28 **(c)** 7 **(d)** 4

23. The number system we use today is called the: 1

(a) Babylonian system **(b)** Maya system
(c) Hindu-Arabic system **(d)** Roman system

24. In an isosceles trapezium ABCD with AB $\parallel$ DC, $\angle A = 65^\circ$. Then $\angle D$ is: 1

(a) 295° **(b)** 115° **(c)** 75° **(d)** 65°

25. Think of 7. Add 5, multiply by 2 and subtract 10. The result is: 1

(a) 9 **(b)** 15 **(c)** 14 **(d)** 19

26. In the expansion of $(2x + 3)(4x + 5)$ the coefficient of x is: 1

(a) 23 **(b)** 22 **(c)** 12 **(d)** 10

27. ₹ 1200 is divided among A, B and C in the ratio 1 : 3 : 4. The smallest share is: 1

(a) ₹ 600 **(b)** ₹ 450 **(c)** ₹ 150 **(d)** ₹ 240

28. The compound interest on ₹ 5000 for 2 years at 10% per annum compounded annually (in ₹) is: 1

(a) 1050 **(b)** 1000 **(c)** 6050 **(d)** 1155

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29. The hypotenuse of a right triangle is 29 cm and one leg is 20 cm. Its area (in cm²) is: 1

(a) 290 **(b)** 230 **(c)** 420 **(d)** 210

30. A car moving at a constant speed covers 210 km in 3 hours. In 8 hours it will cover (km): 1

(a) 560 **(b)** 570 **(c)** 1680 **(d)** 218

31. The top view of a cone standing on its base is a: 1

(a) square **(b)** rectangle

(c) triangle **(d)** circle (with a dot at the centre)

32. The values 5, 6, 7, 10 occur 6, 3, 2, 1 times respectively. The mean is: 1

(a) 6 **(b)** 28 **(c)** 7 **(d)** 3

33. For any n, $(n+1)^2-(n-1)^2$ equals: 1

(a) 4 **(b)** $4n$ **(c)** $2n$ **(d)** 2

34. The area of a trapezium is 70 cm². One parallel side is 12 cm and the height is 7 cm. The other parallel side (in cm) is: 1

(a) 20 **(b)** 8 **(c)** 10 **(d)** 4

35. How many non-square numbers lie between 12^2 and 13^2? 1

(a) 24 **(b)** 23 **(c)** 12 **(d)** 25

36. How many zeros are there in one lakh written in figures? 1

(a) 6 **(b)** 5 **(c)** 4 **(d)** 7

37. Which of the following is NOT a correct Roman numeral? 1

(a) VX **(b)** CM **(c)** XC **(d)** CD

38. Which property holds for every parallelogram? 1

(a) The diagonals are perpendicular **(b)** The diagonals are equal

(c) The diagonals bisect each other **(d)** All sides are equal

39. The smallest digit that can replace * in 63*8 so that the number is divisible by 9 is: 1

(a) 2 **(b)** 5 **(c)** 8 **(d)** 1

40. Which of the following is true for all values of a and b? 1

(a) $(a+b)(a-b)=a^2+b^2$ **(b)** $(a-b)^2=a^2-2ab+b^2$ **(c)** $(a-b)^2=a^2-b^2$ **(d)** $(a+b)^2=a^2+b^2$

41. If the number of workers increases and the job remains the same, the time taken: 1

(a) increases **(b)** stays the same **(c)** decreases **(d)** becomes zero

42. A value changes from 250 to 200. The percentage decrease is: 1

(a) 20% **(b)** 22% **(c)** 18% **(d)** 40%

43. The Baudhayana–Pythagoras theorem states that in a right triangle: 1

(a) the square of a leg equals the sum of the other two sides

(b) the sum of the two legs equals the hypotenuse

(c) the hypotenuse is the shortest side

(d) the square of the hypotenuse equals the sum of the squares of the other two sides

44. Which of the following is the greatest ratio? 1

(a) 3:5 **(b)** 7:12 **(c)** 5:8 **(d)** 4:7

45. A polyhedron has 10 vertices and 15 edges. The number of faces is: 1

(a) 23 **(b)** 7 **(c)** 27 **(d)** 5

46. The median of 18, 49, 17, 5, 24, 11, 20, 40 is: 1

(a) 18 **(b)** 23 **(c)** 19 **(d)** 20

47. Think of a number n. Add 6, multiply by 4, then subtract 24. The result is: 1

(a) $4n$ **(b)** $6n$ **(c)** $4n + 6$ **(d)** $n + 6$

48. The area of a rectangle is 352 cm² and its length is 32 cm. Its breadth (in cm) is: 1

(a) 11 **(b)** 320 **(c)** 176 **(d)** 13

49. The value of $20^2 - 19^2$ is: 1

(a) 39 **(b)** 38 **(c)** 20 **(d)** 40

50. The value of $\dfrac{1}{3^{-2}}$ is: 1

(a) $\frac{1}{9}$ **(b)** -9 **(c)** 9 **(d)** 6

END OF QUESTION PAPER **Total: 50 Marks**

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